

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 1 Lecture - Calculating Gradients

Contact Details

- Dr. Matthew Thomas
- Lecturer in Data Science & Analytics
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About Me

- "Child of the 90s" / "Millenial"
- Mathematician / Statistician
- Work in Environment and Health
- Proud Welshman
- Love TV/Film/Travelling



Plan for Calculus in EART11200

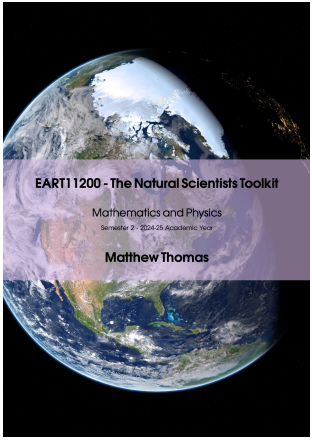
- **Lectures**
  - Monday 15:00-16:00 (Weeks 20–28)
  - Friday 12:00-13:00 (Weeks 29, 33–34)
- Due to issues with timetabling, the room allocations for the lecture change regularly.
- I will place this information on the folders on Blackboard and announce this weekly in the lecture, but please make sure to check timetables to ensure you go to the right place every week.
- You are encouraged to ask questions during the lectures. It is important for me to understand where difficulties may lie.
- If you have any questions beforehand or material you want me to focus on specifically, please email me (matthew.l.thomas@manchester.ac.uk).
- Alternatively, you may submit them anonymously to the Blackboard Discussion Forum.
- You are encouraged to read through slides and look at the weekly exercises beforehand to get the most out of the lectures.

Plan for Calculus in EART11200

- **Practicals**
  - Monday 16:00-18:00 (Weeks 20–28) in Nancy Rothwell 2A.012-2A.013
  - Friday 13:00-15:00 (Weeks 29, 33–34) in Engineering B Blended Theatre 2
- During the first hour, we will work through exercises based on the lecture materials together.
- During the second hour, you will work through exercises which can from the course handbook.
- These sessions are designed to build independence and reinforce the materials (particularly the practical aspects).
- Solutions to problems will be made available alongside the exercises so that you can work through, but you encouraged to have a serious attempt before looking at them.
- There will be teaching assistants in the room alongside me to help you and answer your questions.

Course Handbook

- I have compiled a Course Handbook for Calculus.
- This contains supporting notes, as well as weekly exercises and their solutions.
- The exercises are designed to aid your understanding of the tools we learn in this course. These are the ones you will work through in the practicals.
- Also contains a course crib sheet, equations presented in the exam, some useful formulae and physical units.
- The handbook is not designed to replace your own notes. You should make your own notes in the lectures as the act of writing down notes/equations is a good way to learn.



Office Hours and Feedback

- I will not be running a formal office hour due to the number of students.
- If you feel like you need help outside of the timetabled sessions, then please do email me (matthew.l.thomas@manchester.ac.uk) and we can book something in if we can't resolve it by email.
- I'm very open to receiving feedback during the course of the module, and will do my best to accommodate requests and fix issues.
- You can provide this either in-person at the timetabled sessions or by email.
- You can also provide feedback on the module assessment form when this gets released.

What to expect from us and what we expect from you

- Try to solve problems before you get a better feel for your ability
- Everyone is at a different stage – it is ok to feel frustrated, that's life.
- Use any frustration to improve. Step back, then try again.
- Give it your best shot!
- Ask for help - from lecturers / teaching assistant / peers
- Practice is the only way forward

You CAN and WILL improve your mathematical ability by following the above.

Course Intended Learning Outcomes

(1) practice the application of key concepts in maths, physics and chemistry, including manipulating logarithms, rearranging equations, algebra, statistics, basic calculus, geometry, and atomic and molecular structure.

(2) apply mathematical theory to break down and model / describe real-world example problems.

(3) use calculus to solve ordinary and partial differential equations by separation of variables, and to arrive at approximate solutions using finite-difference.

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

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Intended Learning Outcomes

(1) Discuss the role of calculus in broader mathematical and physical contexts.

(2) Apply Euler's Backward Method to analyse rates of change in mathematical functions and real world applications.

(3) Apply Euler's Forward Method to analyse rates of change in mathematical functions and real world applications.

(4) Apply Central Differences Method to analyse rates of change in mathematical functions and real world applications.

WHY?

If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generation of creatures, what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis that all things are made of atoms — **little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another**. In that one sentence, you will see, there is an enormous amount of information about the world, if just a little imagination and thinking are applied.



Richard Phillips Feynman
May 11, 1918 – February 15, 1988
American theoretical physicist

WHY?

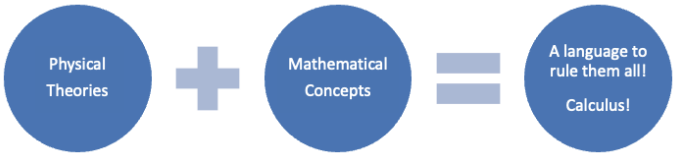


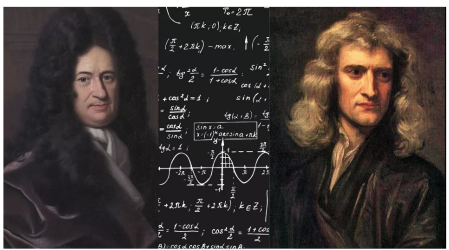
The iconic 'Blue Marble' image taken from Apollo 17, 1972

How about something a little more dynamic...

`https://www.youtube.com/watch?v=oRsY_UviBPE`

What is Calculus?





- Calculus gives us the tools to calculate how quickly something is changing for example, or using that information to calculate quantities and states in the past or future.
- Is that relevant to understanding our changing climate?

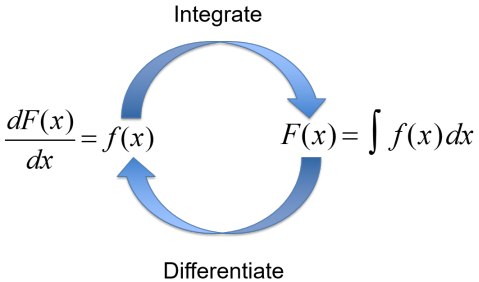
What is Calculus?

- Calculus allows us to investigate drivers behind:
  - Climate change
  - Air quality
  - Species population dynamics
  - Environmental engineering
  - Plate tectonics
  - any more?!
- **Confidence, of all levels, in quantitative analysis/prediction is important!**

What is Calculus?

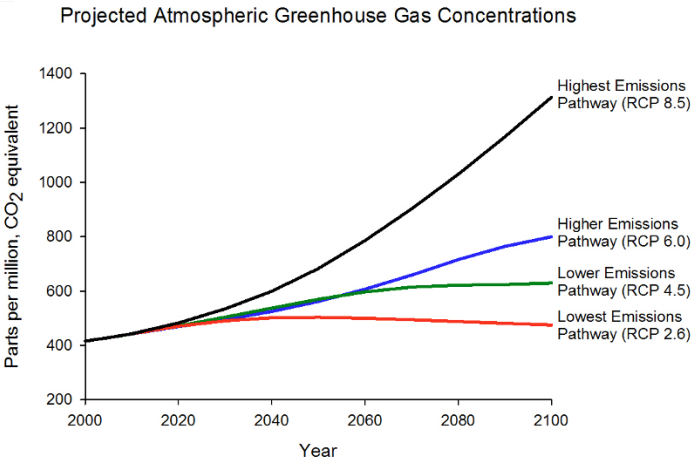
- Its is also a language and a way of reasoning. In this course we will learn to use specific notation when we construct that reasoning and solve several physical laws.

e.g. An equation that calculates the rate of change of temperature in the past, present or future



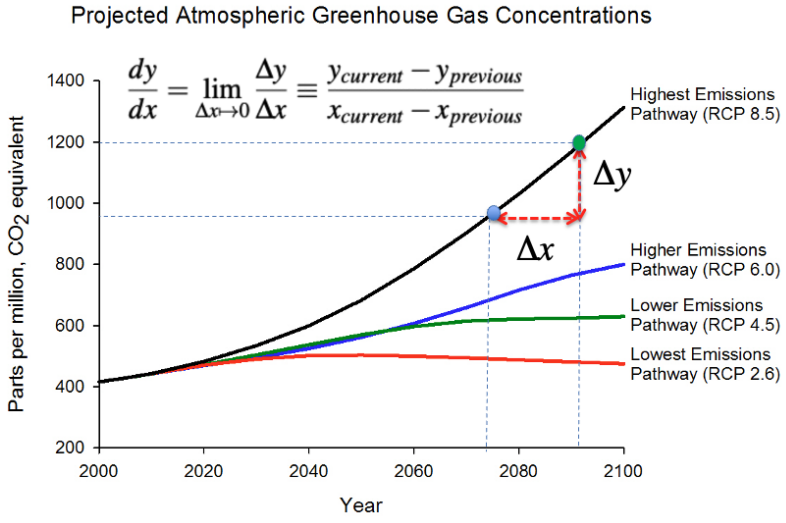
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Calculating Gradients

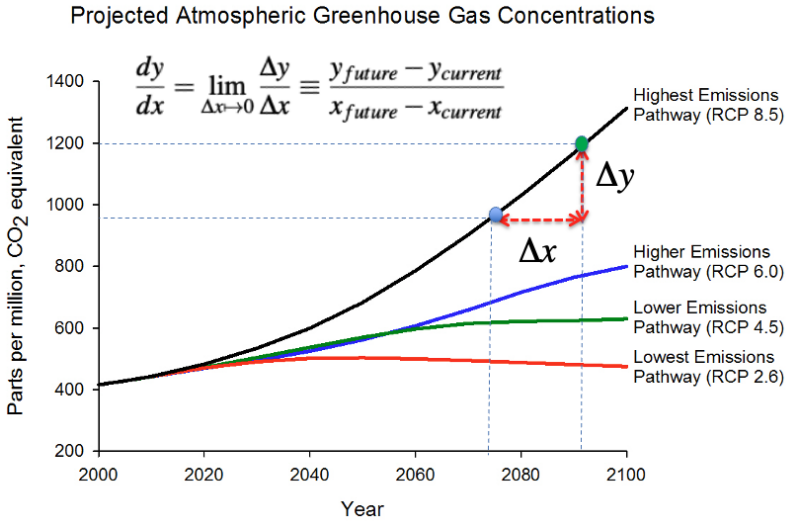


- https://www.epa.gov/climate-change-science/future-climate-change

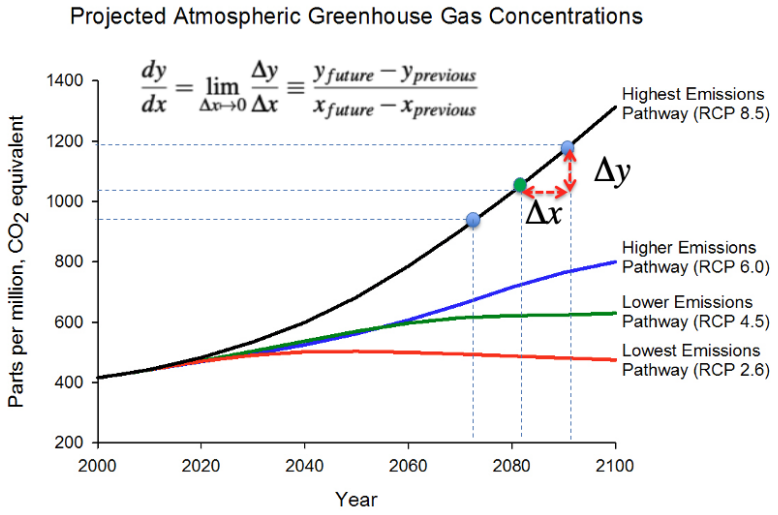
Calculating Gradients: Euler's Backward Method



Calculating Gradients: Euler's Forward Method



Calculating Gradients: Central Differences Method

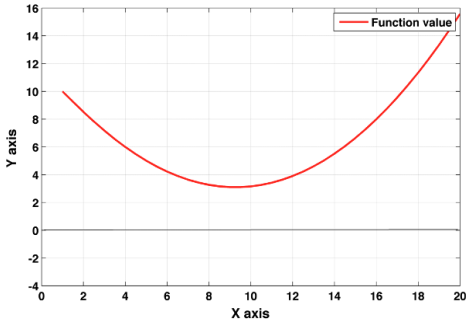


Calculating Gradients

- For the function, can we find the gradient at $x = 4$?

$$y = \sqrt{x} + \left(\frac{x - 10}{3}\right)^2$$

- Do you use future values, previous or both??



$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_x - y_{x-\Delta x}}{\Delta x}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_x}{\Delta x}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$

Differentiation: How do we do it?

- For a function, $y = f(x)$, that describes some process we are interested in, in one dimension, the derivative:

$$f'(x) = \frac{dy}{dx}$$

- It is interpreted as the rate of change of 'y' with respect to 'x'. Lets imagine we are observing this change over a discrete interval, such that we could write this as an approximation:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

- There are rules that enable us to use an analytical solution, rather than numerical…more on this next week!

Calculating Gradients

- For the function, can we find the gradient at $x = 4$?

$$y = \sqrt{x} + \left(\frac{x - 10}{3}\right)^2$$

- We see that using a step-size of $\Delta x = 0.25$ that the estimated gradient:
  - Euler's Backward Method: $\frac{\Delta y}{\Delta x} = -1.1071$
  - Euler's Forward Method: $\frac{\Delta y}{\Delta x} = -1.0593$
  - Central Differences Method: $\frac{\Delta y}{\Delta x} = -1.0832$
- The analytical derivative happens to be given by the following:

$$\frac{dy}{dx} = 0.5x^{-0.5} + \frac{2}{3}\left(\frac{x - 10}{3}\right)$$

- At $x = 4$, $\frac{dy}{dx}(4) = -1.08333$ so we can estimate the accuracy of each method!